Finite-Core Dynamo Hypothesis and Shell-Current Dynamics

Marcin Słowik
ORCID: 0009-0004-6707-4546
Zielona Góra

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Abstract

This work formalizes the finite-core dynamo hypothesis, integrating the tree-trunk structure with shell-current dynamics ($\mathbf{J}_{\text{shell}}$), relativistic corrections, and energy balance. By enforcing the zero-flux condition $B_n = 0$ and tracking magnetar-scale bursts via tree shakes, we unify compact object evolution with the Package Formula ($1\,P = 2.8\,M_\odot$).

1 Shell-Current Dynamics and Relativistic Corrections

In the finite-core model, the surface current density in the outer shell of the tree trunk $\mathbf{J}_{\text{shell}}$ arises directly from the rotation of degenerate conductive matter around the dense inner core. We express this in a relativistic form, accounting for the local rotational velocity of the shell v approaching the speed of light c:

$$\mathbf{J}_{\text{shell}} = \sigma \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} \right)$$

Where σ denotes the electrical conductivity of the medium, and the magnetic field $\mathbf{B}$ generates magnetic pressure balancing local stresses within the shell. The stability condition of the trunk, $B_n = 0$, dictates that the total magnetic flux in the equilibrium state is quantized and confined within the structure:

$$\oint_{\partial V_{\text{shell}}} \mathbf{B} \cdot d\mathbf{S} = 0$$

2 Tree Shakes and Magnetar Bursts

When additional matter accretes from the accretion disk, a fresh portion of momentum and charge is delivered to the shell. This causes a local violation of the $B_n = 0$ condition:

$$\Delta E_{\text{mag}} = \int_{V_{\text{shell}}} \left(\frac{(B + \delta B)^2 - B^2}{2\mu_0} \right) dV > 0$$

At this point, the maintained equilibrium $S_n = E_{\text{core}} + E_{\text{magnetic}} = 2F_n$ is temporarily disrupted. The system reacts violently through a tree shake, instantly emitting the excess magnetic energy as a powerful burst:

$$E_{\text{flash}} = \Delta E_{\text{mag}} \quad \xrightarrow{\text{instant relaxation}} \quad B_n = 0$$

3 Energy Balance and the Package Formula

The total energy balance linking the dense center and the magnetic shell is given by:

$$S_n = 2F_n = E_{\text{core}} + E_{\text{magnetic}}$$

Where the magnetic energy of the shell is:

$$E_{\text{magnetic}} = \left(\frac{B^2}{2\mu_0}\right) \cdot V_{\text{shell}}$$

For a typical star of mass $1.4\,M_\odot$ and core radius 10 km, with a shell thickness of 1 km ($V_{\text{shell}} \approx 10^{12}$ m^3), achieving a magnetic field of 10^{15} G (10^{11} T, magnetar scale) requires consuming only about 0.00003% of the total gravitational collapse energy ($E_{\text{grav}} \approx \frac{GM^2}{R} \approx 3 \times 10^{46}$ J).

Each subsequent mass packet ($1\,P = 2.8\,M_\odot$) entering the macroscopic structure ($R_s = N \times 8.27$ km) carries a specific momentum and rotation budget that directly feeds the shell's magnetic field while preserving $B_n = 0$.